

GII Index Reduction Analysis Report

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1 Executive Summary

This report documents a method for determining how far the number of indicators in the Global Innovation Index (GII) 2025 can be reduced without significantly compromising the index’s discriminatory power. The method was previously applied to the EU JRC’s Sustainable and Inclusive Well-being Index (Benczur et al. 2024; Boskovic et al. 2024); this report replicates it on GII’s 78 Level-1 indicators observed for 139 countries/economies.

Two perspectives are evaluated: a **country perspective** (do countries end up the same distance apart using a subset as with the full indicator set?) and a **variable perspective** (does a subset preserve the PCA structure of the full set?). The two perspective rely on complementary metrics. The country perspective is based on a Spearman rank correlation between full- and subset-distance matrices; the variable perspective is based on r_m^2 - the percentage of the full PCA’s variance structure recovered by the subset.

Both perspective require the analyst to choose a sub-set of indicators. We have opted for three strategies here. All of which are based on either 21 or 42 indicators. As the GII has 21 sub-pillars, it is the minimum needed to cover all the concepts it aims to measures. The strategy to select the subsets is to find the best sub-set from 1000 random selections. This method represents a baseline which ignores the conceptual structure. The second strategy is to select one indicator per-pillar in an attempt to follow the conceptual structure at the most detailed level. The third strategy is to proportionally allocate indicators to pillars, with the idea that the number of indicators within a pillar also reflect it’s importance or difficulty of measurement.

The the headline recommendation is that the “one indicator per sub-pillar” set of **21- indicators comes close to reproducing the PCA structure** ($r_m = 0.826$, i.e. $r_m^2 = 68.2\%$ of the full PCA’s variance structure) while also closely reproducing the distance between countries (Rand

Index 0.923 against the full-data country clustering). A larger 42-indicator subset, built by allocating indicators proportionally to each pillar’s actual size rather than a flat count per pillar, reaches $r_m^2 = 92.6\%$ and a rank correlation of 0.941 with the full country-distance matrix. Both are documented below alongside the flat and proportional strategies they were chosen from.

At the 21-indicator budget, exhaustive within-group search does not clearly outperform a random subset of similar size, for either metric; the gap becomes more visible at the larger 42-indicator budget. Both perspectives’ results below discuss this directly.

2 Introduction

When creating indicator dashboards and composite indices, indicator selection can be daunting. Stakeholders, statistics and conceptual relevance often diverge in recommendations or build up a large set of indicators that is impractical. A key statistical question is: how much of the information content (“variation”) of the full indicator set is preserved in a candidate subset?

This analysis applies two methodological approaches to the GII 2025 data (78 Level-1 indicators, 139 countries):

1. **Country Perspective:** Assessing how similar countries are using all indicators versus a subset, using distance matrices and clustering techniques
2. **Variable Perspective:** Using Principal Component Analysis (PCA) to measure how much information a subset captures compared to the full set

2.1 Finding a suitable reduced set

Composite indices such as the GII combine two different measurement logics. At the upper levels - pillars, sub-indices, the index itself - the structure is *formative*: each pillar captures a distinct dimension of innovation performance, and the index’s meaning depends on all of them being represented; dropping a pillar entirely would change what the index measures, not just how precisely it measures it. Within a sub-pillar, by contrast, the indicators are closer to *reflective* measures of a single underlying concept - largely interchangeable manifestations of the same thing, where one or two well-chosen indicators can stand in for the group without changing what is being measured.

This distinction motivates the reduction strategy used throughout this analysis: every pillar and sub-pillar is retained in every candidate subset, preserving the formative structure and therefore the index’s meaning, while indicator reduction happens only *within* each group, by selecting which reflective indicator(s) best represent it. We apply two strategies to decide *how many* reflective indicators to draw from each formative group. The first is to select one indicator per sub-pillar and the second is to select a number per pillar proportional to it’s original size.

The metrics for these two strategies ignore correlations between pillars or sub-pillars. However, these cannot be incorporated without also expanding the search to be exhaustive.

We have not implemented an exhaustive search for the best subset. Even restricted to searching *within* a single pillar, exhaustive search grows quickly. Selecting the best possible subset of size k out of p indicators requires evaluating $C(p, k) = \frac{p!}{k!(p-k)!}$ combinations. The Institutions pillar (IN.1) has 6 indicators, so choosing 2 of them already means 15 possible combinations. Multiplied across all 7 pillars, the number of combinations for a subset is prohibitive. This analysis searches exhaustively within each pillar or sub-pillar rather than across the whole index at once. The table below shows how many combinations remain even after allocating a 20-indicator budget proportionally across pillars.

Table 1: Unique combinations per pillar of GII when allocating a 20-indicator budget proportionally

Pillar	Indicators in Pillar	Allocated (budget=20)	Combina- tions
Institutions	6	1	6
Human capital and research	12	3	220
Infrastructure	9	2	36
Market sophistication	11	3	165
Business sophistication	15	4	1365
Knowledge and technology outputs	14	4	1001
Creative outputs	11	3	165

Even after this proportional allocation collapses the problem to one pillar at a time, there are still 1,767,710,124,180,000 combinations to explore in total. Underscoring why within-group exhaustive search is the only tractable approach.

3 Methodology

3.1 Statistical Methods Explanation

The variable perspective uses a subset-selection approach based on Principal Component Analysis (PCA). Rather than simply ranking individual variables by importance, this method evaluates entire subsets of variables according to how well they reproduce the PCA structure of the full dataset.

The criterion, r_m , was introduced by McCabe (1984) as one of several optimality criteria for selecting “principal variables”. In contrast to principle components, principle variables are a

subsets of the original variables that best capture variation in the original set. The method was developed in order to select indicators which could function as a principle component while remaining interpretable. Cadima and Jolliffe (2001) formalize this same criterion as one of two indicators for how well a subset of variables approximates a full PCA, and show that r_m^2 is exactly the percentage of total variance in the original data accounted for by regressing the full data on the chosen subset. This is the interpretation of r_2^m used throughout this report. Both papers show, that choosing variables by their PC loadings alone frequently produces subsets that are suboptimal compared to subsets chosen by directly maximizing r_m .

A key advantage of this method is its built-in protection against redundancy. For example, if multiple indicators are highly correlated, including one in the subset provides most of the information, and adding the others contributes relatively little additional ability to reconstruct the principal components.

The mathematical formula for the selection criterion is:

$$r_m = \sqrt{\frac{\sum_{i=1}^p \lambda_i r_{mi}^2}{\sum_{j=1}^p \lambda_j}}$$

where: - λ_i is the eigenvalue of principal component i (PC_i) from the full dataset, representing the variance explained by that component. The subscript m indexes a candidate subset - r_{mi}^2 is the coefficient of determination (R^2) between principal component i and the candidate subset of variables. Since PC_i is a linear combination of the original variables, the R^2 of the subset measures how well the principal component can be reconstructed from the subset.

Conceptually, this formula computes a weighted average of how well the subset predicts each principal component, with weights given by the variance explained by each component, then normalizes by the total variance. The resulting r_m^2 value can be interpreted as the percentage of total variation explained by the subset, providing an intuitive measure of information retention.

This method assumes that “information” is defined by the variance structure captured by PCA, which is appropriate for statistical data reduction. However, it may downweight variables with co-variance but high policy relevance, which is an important consideration for substantive index development (c.f.(Algursan et al. 2026)).

3.1.1 The Country Perspective’s Distance-Based Criterion

The country perspective asks assess whether a subset preserves how similar countries are judged to be relative to one another. For each candidate subset, the pairwise Euclidean distance between every pair of countries is computed using only that subset’s indicators, and the ranking of those pairwise distances is compared to the ranking obtained from the full indicator set, via

Spearman rank correlation. A correlation near 1 means the subset reproduces almost the same country-similarity ordering as the full index.

3.2 Proportional Allocation Methodology

Both perspectives select candidate subsets by exhaustive search *within* each pillar or sub-pillar, then assemble the per-group picks into a full subset - consistent with the formative/reflective distinction. Every pillar and sub-pillar is represented in every candidate subset, and only the reflective indicators within each group are subject to selection. This requires deciding how many indicators to take from each group. Two allocation rules are used throughout this report:

- **Flat allocation of 21 indicators:** every pillar contributes (3) indicators, or every sub-pillar contributes 1 indicators. A simple strategy to implement, however, it ignores that GII's pillars differ in the number of indicators, from 6 indicators (Institutions) to 15 (Business sophistication), and its sub-pillars range from 2 to 5 indicators. A flat allocation over-represents small groups and under-represents large ones relative to their actual share of the index. This may limit the subset's utility statistically and leave out conceptual nuances of the GII measurement framework.
- **Proportional allocation of 21 or 42 indicators.** Each group's budget is set proportional to its share of all indicators, using a **largest-remainder (Hamilton) apportionment**. Every group first receives n indicators rounded down to the lowest integer. Any indicators left over are handed out one at a time to the groups with the largest fractional remainder. Unlike flat allocation, a group with a very small share of the index can receive **zero** indicators under this rule. The total budget used is **42** indicators (double the flat allocations' 21). For proportional budget to be different from flat allocation we needed to increase the budget to be greater than the number of sub-pillars.

In the case of the variable perspective, the indicators are chosen by exhaustive search maximizing a *local* r_m or a local proxy computed the same way (for the country perspective) - restricted to that group's own sub-PCA rather than the full 78-indicator PCA. This local search is a tractable stand-in for an otherwise infeasible full search. The assembled subset is then re-scored against the full-dataset to obtain the metrics reported in this document.

3.3 Data Description

The Global Innovation Index (GII) 2025 dataset ((WIPO) 2024) comprises 78 Level-1 indicators observed for 139 countries/economies, organized in a 5-level hierarchy: 78 indicators -> 21 sub-pillars -> 7 pillars -> 2 sub-indices (Inputs, Outputs) -> 1 overall index. Indicators with negative polarity (e.g. pupil-teacher ratio, applied tariff rate) are reversed before standardization, so that higher standardized values consistently indicate better innovation performance across all indicators. Missing values are imputed via 5-nearest-neighbors using the VIM (Kowarik and Templ 2016) package prior to z-scoring; see the Appendix for imputation statistics.

4 Key Findings

4.1 Country Perspective

4.1.1 Distance Matrix Analysis of Random Subsets

We selected random subset sizes ranging from 10% to 90% of all 78 indicators, while also selecting specifically `CONFIG$exhaustive$proportional_target`, matching the proportional strategies’ budget below. Each k subset was drawn 1000 times and compared to the full-dataset Euclidean distance matrix via Spearman rank correlation:

Table 2: Rank correlation of random subsets vs. the full distance matrix, by subset size k

Subset Size (k)	Mean Correlation	Median Correlation	Min Correlation	Max Correlation
8	0.678	0.687	0.383	0.824
15	0.784	0.789	0.626	0.878
23	0.847	0.851	0.726	0.918
31	0.891	0.893	0.808	0.936
38	0.920	0.922	0.856	0.955
42	0.932	0.935	0.872	0.962
46	0.946	0.948	0.898	0.971
54	0.963	0.965	0.902	0.983
62	0.978	0.979	0.945	0.991
69	0.988	0.989	0.957	0.996

Correlation rises smoothly and near-monotonically with subset size, from a mean of 0.678 at $k = 8$ to 0.988 at $k = 69$. This random-subset curve is the baseline to the local exhaustive-search strategies.

4.1.2 Flat Allocation

Table 3: Best 3 indicators per pillar, by rank correlation with the full distance matrix

Pillar	r_m of Best Subset	Selected Indicators
Institutions	0.965	Entrepreneurship policies and culture†; Policy stability for doing business†; Regulatory quality*
Human capital and research	0.757	Expenditure on education; Gross expenditure on R&D; Tertiary enrolment

Pillar	r_m of Best Subset	Selected Indicators
Infrastructure	0.784	Gross capital formation; ICT access*; Low-carbon energy use
Market sophistication	0.764	Domestic credit to private sector; Domestic market scale; Joint venture deals
Business sophistication	0.746	Females employed w/advanced degrees; High-tech imports; University–industry R&D collaboration†
Knowledge and technology outputs	0.702	Software spending; Production and export complexity; Patents by origin
Creative outputs	0.697	Entertainment and media market; National feature films; Industrial designs by origin

Table 4: Best indicator per sub-pillar, by rank correlation with the full distance matrix

Sub-Pillar	r_m of Best Subset	Selected Indicators
Institutional environment	0.938	Operational stability for businesses*
Regulatory environment	0.975	Rule of law*
Business environment	0.834	Policy stability for doing business†
Education	0.626	School life expectancy
Tertiary education	0.615	Tertiary enrolment
Research and development (R&D)	0.793	Gross expenditure on R&D
Information and communication technologies (ICTs)	0.877	ICT access*
General infrastructure	0.647	Logistics performance*
Ecological sustainability	0.588	ISO 14001 environmental certificates
Credit	0.650	Finance for startups and scale-ups†
Investment	0.750	Joint venture deals
Trade, diversification and market scale	0.639	Domestic industry diversification
Knowledge workers	0.802	Knowledge-intensive employment
Innovation linkages	0.727	University–industry R&D collaboration†
Knowledge absorption	0.508	Research talent
Knowledge creation	0.638	PCT patents by inventor origin

Sub-Pillar	r_m of Best Subset	Selected Indicators
Knowledge impact	0.592	High-tech manufacturing
Knowledge diffusion	0.649	Production and export complexity
Intangible assets	0.613	Industrial designs by origin
Creative goods and services	0.550	Entertainment and media market
Online creativity	0.639	GitHub commits

The selection of (21 indicators by flat allocation of 3 indicators per pillar reaches a correlation of 0.834, while (21 indicators (1/sub-pillar) reaches 0.844. Both are close to the random-subset baseline at a comparable size (mean 0.847 at $k = 23$ in the table above), suggesting that at a 21-indicator budget, exhaustive within-group search buys little over chance for this metric. However, the method is guaranteed to preserve the conceptual structure and therefore more likely to preserve the meaning of the formative indicator.

4.1.3 Clustering Results: SIMPROF + Rand Index

Countries were clustered via SIMPROF (significance-tested hierarchical clustering, see Clarke et al. (2008)) on the full 78-indicator dataset and separately compared via Rand Index. The Rand Index for flat allocation by sub-pillar (1 indicator per sub-pillar) was 0.923, while flat allocation by pillar (3 indicators per pillar) yielded 0.917.

Table 5: Rand Index: full-data clustering vs. flat-allocation clustering (by pillar and by sub-pillar)

Comparison	Rand Index	Full Indicator Count	Subset Indicator Count	Allocation Rule
Full vs. Flat allocation (sub-pillar)	0.923	78	21	Flat allocation (sub-pillar)
Full vs. Flat allocation (pillar)	0.917	78	21	Flat allocation (pillar)

4.1.4 Proportional Allocation

We proportionally allocate 42 indicators, using the largest-remainder method described in Methodology above.

Table 6: Proportional allocation across pillars

Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Institutions	3	0.965	Entrepreneurship policies and culture†; Policy stability for doing business†; Regulatory quality*
Human capital and research	6	0.898	Expenditure on education; Global corporate R&D investors; Researchers; Graduates in science and engineering; Tertiary enrolment; Tertiary inbound mobility
Infrastructure	5	0.900	GDP/unit of energy use; Gross capital formation; ICT access*; ISO 14001 environmental certificates; Low-carbon energy use
Market sophistication	6	0.926	Applied tariff rate, weighted avg.; Domestic industry diversification; Joint venture deals; Late VC deals intensity; Market capitalization; Loans from microfinance institutions
Business sophistication	8	0.932	Emerging knowledge workforce; FDI net inflows; GERD performed by business; High-tech imports; ICT services imports; Intellectual property payments; Patent families; State of cluster development†
Knowledge and technology outputs	8	0.915	Software spending; Labor productivity growth; High-tech exports; ICT services exports; ISO 9001 quality certificates; PCT patents by inventor origin; Unicorn valuation; Utility models by origin
Creative outputs	6	0.865	Mobile app creation; Intangible asset intensity, top 15; Creative goods exports; Entertainment and media market; National feature films; Trademarks by origin

Table 7: Proportional allocation across sub-pillars

Sub-Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Institutional environment	1	0.938	Operational stability for businesses*
Regulatory environment	1	0.975	Rule of law*
Business environment	1	0.834	Entrepreneurship policies and culture†

Sub-Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Education	3	0.882	Expenditure on education; PISA scales in reading, maths and science; Pupil-teacher ratio, secondary
Tertiary education	2	0.840	Graduates in science and engineering; Tertiary inbound mobility
Research and development (R&D)	2	0.922	Global corporate R&D investors; Researchers
Information and communication technologies (ICTs)	2	0.966	Government's online service; <i>ICT use</i>
General infrastructure	2	0.867	Electricity output; Gross capital formation
Ecological sustainability	1	0.588	ISO 14001 environmental certificates
Credit	1	0.650	Finance for startups and scale-ups†
Investment	3	0.980	Joint venture deals; Late VC deals intensity; Market capitalization
Trade, diversification and market scale	1	0.639	Domestic industry diversification
Knowledge workers	3	0.944	Females employed w/advanced degrees; GERD financed by business; GERD performed by business
Innovation linkages	3	0.937	Patent families; Public research–industry co-publications; State of cluster development†
Knowledge absorption	3	0.813	High-tech imports; ICT services imports; Intellectual property payments
Knowledge creation	3	0.907	PCT patents by inventor origin; Scientific and technical articles; Utility models by origin
Knowledge impact	2	0.777	Labor productivity growth; High-tech manufacturing
Knowledge diffusion	3	0.871	High-tech exports; Intellectual property receipts; ISO 9001 quality certificates
Intangible assets	2	0.811	Global brand value, top 5,000; Trademarks by origin
Creative goods and services	2	0.739	Creative goods exports; National feature films
Online creativity	1	0.639	GitHub commits

Finding: At the same 42-indicator budget, pillar level proportional allocation achieves a rank correlation of 0.941, compared to 0.929 sub-pillar-level.

Table 8: Rand Index: full clustering vs. proportional-allocation subset clustering

Comparison	Rand Index	Full Indicator Count	Subset Indicator Count	Allocation Rule
Full vs. Proportional allocation (pillar)	0.885	78	42	Proportional allocation (pillar)

Finding: The 42-indicator proportional subset recovers a Rand Index of 0.885 - notably *lower* than flat allocation of 21 indicator’s Rand Index (0.923). Despite covering double the indicators and scoring higher on rank correlation. Rand Index is not monotonic in subset size or in rank correlation: it depends on where SIMPROF’s significance testing happens to draw cluster boundaries, which can shift disproportionately with a different indicator mix. At the 42-indicator budget, a *random* subset already averages a rank correlation of 0.932 (Fig 2 above) - only marginally below proportional allocation of 21 indicators rank correlation of 0.941. Exhaustive search is not clearly outperforming chance at this budget either, echoing the pattern seen at the smaller 21 indicator budget above.

4.2 Variable Perspective

4.2.1 PCA Analysis

The table below shows the number of principal components needed to capture specified levels of variance in the full 78-indicator dataset:

Table 9: Components needed for target variance levels (full PCA)

Variance Level	Components Needed
80%	17
90%	26
95%	36

Finding: The first 17 of 78 components already capture 80% of total variance, confirming substantial redundancy in the full indicator set - a necessary condition for indicator reduction to be viable at all. Note this is a property of the full PCA (unconstrained linear combinations of all 78 indicators), not directly comparable to how many *raw indicators* a subset needs - see the r_m vs. #PCs comparison below.

4.2.2 Reflective Indicators per Sub-Pillar (Flat Allocation - Headline Subset)

Per the project’s specified methodology, the headline recommended subset chooses one variable per sub-pillar (Level 2 aggregate), selecting within each sub-pillar the indicator with the best local r_m (“reflective indicator” - the one that best represents its sub-pillar’s own variance structure):

Table 10: Flat allocation (sub-pillar): reflective indicator per sub-pillar

Sub-Pillar	r_m of Best Subset	Selected Indicators
Institutional environment	0.938	Operational stability for businesses*
Regulatory environment	0.975	Rule of law*
Business environment	0.834	Entrepreneurship policies and culture†
Education	0.626	School life expectancy
Tertiary education	0.615	Tertiary enrolment
Research and development (R&D)	0.793	Gross expenditure on R&D
Information and communication technologies (ICTs)	0.877	ICT access*
General infrastructure	0.647	Logistics performance*
Ecological sustainability	0.588	ISO 14001 environmental certificates
Credit	0.650	Finance for startups and scale-ups†
Investment	0.750	Joint venture deals
Trade, diversification and market scale	0.639	Domestic industry diversification
Knowledge workers	0.802	Knowledge-intensive employment
Innovation linkages	0.727	University–industry R&D collaboration†
Knowledge absorption	0.508	Research talent
Knowledge creation	0.638	PCT patents by inventor origin
Knowledge impact	0.592	High-tech manufacturing
Knowledge diffusion	0.649	Production and export complexity
Intangible assets	0.613	Industrial designs by origin
Creative goods and services	0.550	Entertainment and media market
Online creativity	0.639	GitHub commits

Finding: This 21-indicator subset achieves $r_m = 0.826$, i.e. $r_m^2 = 68.2\%$ of the full dataset’s PCA variance structure.

4.2.3 Flat Allocation by Pillar (3 per Pillar) - Comparison

Table 11: Flat allocation (pillar): best 3 indicators per pillar

Pillar	r_m of Best Subset	Selected Indicators
Institutions	0.965	Entrepreneurship policies and culture†; Policy stability for doing business†; Regulatory quality*
Human capital and research	0.757	Expenditure on education; Gross expenditure on R&D; Tertiary enrolment
Infrastructure	0.784	Gross capital formation; ICT access*; Low-carbon energy use
Market sophistication	0.764	Domestic credit to private sector; Domestic market scale; Joint venture deals
Business sophistication	0.746	Females employed w/advanced degrees; High-tech imports; University–industry R&D collaboration†
Knowledge and technology outputs	0.702	Software spending; Production and export complexity; Patents by origin
Creative outputs	0.697	Entertainment and media market; National feature films; Industrial designs by origin

Finding: Flat allocation by pillar reaches $r_m = 0.854$ ($r_m^2 = 72.9\%$) - marginally *higher* than flat allocation by sub-pillar’s 0.826 at the same 21-indicator size. Flat allocation by sub-pillar is nonetheless used as the headline recommendation, not flat allocation by pillar, because the project’s methodology specifically calls for “one variable per level-2 aggregate, justified by reflective indicator” rather than whichever flat allocation happens to score marginally higher - the sub-pillar-level selection is the specified deliverable, and the score difference between the two is small enough not to override that.

4.2.4 Proportional Allocation by Pillar and Sub-Pillar - Comparison

Table 12: Proportional allocation across pillars

Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Institutions	3	0.965	Entrepreneurship policies and culture†; Policy stability for doing business†; Regulatory quality*
Human capital and research	6	0.898	Expenditure on education; Global corporate R&D investors; Researchers; Graduates in science and engineering; Tertiary enrolment; Tertiary inbound mobility
Infrastructure	5	0.900	GDP/unit of energy use; Gross capital formation; ICT access*; ISO 14001 environmental certificates; Low-carbon energy use
Market sophistication	6	0.926	Applied tariff rate, weighted avg.; Domestic industry diversification; Joint venture deals; Late VC deals intensity; Market capitalization; Loans from microfinance institutions
Business sophistication	8	0.932	Emerging knowledge workforce; FDI net inflows; GERD performed by business; High-tech imports; ICT services imports; Intellectual property payments; Patent families; State of cluster development†
Knowledge and technology outputs	8	0.915	Software spending; Labor productivity growth; High-tech exports; ICT services exports; ISO 9001 quality certificates; PCT patents by inventor origin; Unicorn valuation; Utility models by origin
Creative outputs	6	0.865	Mobile app creation; Intangible asset intensity, top 15; Creative goods exports; Entertainment and media market; National feature films; Trademarks by origin

Table 13: Proportional allocation across sub-pillars

Sub-Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Institutional environment	1	0.938	Operational stability for businesses*
Regulatory environment	1	0.975	Rule of law*
Business environment	1	0.834	Entrepreneurship policies and culture†

Sub-Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Education	3	0.882	Expenditure on education; PISA scales in reading, maths and science; Pupil-teacher ratio, secondary
Tertiary education	2	0.840	Graduates in science and engineering; Tertiary inbound mobility
Research and development (R&D)	2	0.922	Global corporate R&D investors; Researchers
Information and communication technologies (ICTs)	2	0.966	Government's online service; <i>ICT use</i>
General infrastructure	2	0.867	Electricity output; Gross capital formation
Ecological sustainability	1	0.588	ISO 14001 environmental certificates
Credit	1	0.650	Finance for startups and scale-ups†
Investment	3	0.980	Joint venture deals; Late VC deals intensity; Market capitalization
Trade, diversification and market scale	1	0.639	Domestic industry diversification
Knowledge workers	3	0.944	Females employed w/advanced degrees; GERD financed by business; GERD performed by business
Innovation linkages	3	0.937	Patent families; Public research–industry co-publications; State of cluster development†
Knowledge absorption	3	0.813	High-tech imports; ICT services imports; Intellectual property payments
Knowledge creation	3	0.907	PCT patents by inventor origin; Scientific and technical articles; Utility models by origin
Knowledge impact	2	0.777	Labor productivity growth; High-tech manufacturing
Knowledge diffusion	3	0.871	High-tech exports; Intellectual property receipts; ISO 9001 quality certificates
Intangible assets	2	0.811	Global brand value, top 5,000; Trademarks by origin
Creative goods and services	2	0.739	Creative goods exports; National feature films
Online creativity	1	0.639	GitHub commits

Finding: At 42 indicators, proportional allocation by pillar reaches $r_m = 0.963$ ($r_m^2 = 92.6\%$) and proportional allocation by sub-pillar reaches $r_m = 0.946$ ($r_m^2 = 89.5\%$) - both a large step up from the 21-indicator flat allocations, as expected with double the budget.

4.2.5 Flat-Allocation Sweep: 1-4 Indicators per Pillar

To see how r_m grows as the flat per-pillar allocation increases, the flat allocation by pillar search was repeated for 1, 2, 3, and 4 indicators per pillar:

Table 14: r_m as the flat per-pillar allocation increases

Indicators per Pillar	r_m	Total Indicators
1	0.705	7
2	0.802	14
3	0.854	21
4	0.900	28

Finding: r_m rises from 0.705 at 7 variables (1/pillar) to 0.9 at 28 variables (4/pillar) - a steep gain per additional variable early on, consistent with the PCA finding above that a small number of dimensions already captures most of the variance structure.

4.2.6 Random-Subset r_m Curve vs. #PCs Needed

For the same subset sizes used in the country perspective's Fig 2, r_m was computed for 20 random subsets per size. r_m is far more expensive per evaluation than the rank-correlation metric - it requires regressing all 78 principal components on the candidate subset - so fewer random draws were used per size here than Fig 2's 1,000.

Table 15: Random-subset r_m by size, vs. cumulative variance explained by the first k principal components

Subset Size (k)	r_m	Subset Variance Captured	PCA Variance (k components)
8	0.695	0.484	0.649
15	0.791	0.626	0.785
23	0.857	0.734	0.878
31	0.900	0.811	0.931
38	0.925	0.856	0.959
46	0.951	0.904	0.978
54	0.969	0.940	0.989
62	0.984	0.968	0.996

Subset Size (k)	r_m	Subset Variance Captured	PCA Variance (k components)
69	0.991	0.983	0.999

Finding: At $k = 21$ (interpolating between $k = 15$ and $k = 23$ above), a random subset averages roughly the same r_m as the exhaustive flat-allocation results report (0.854 by pillar / 0.826 by sub-pillar) - the same pattern seen in the country perspective, where exhaustive within-group search does not clearly beat chance at this budget. The PCA-variance column is consistently higher than a same-sized *indicator* subset's r_m^2 at every k - expected, since the first k principal components are unconstrained linear combinations of all 78 indicators, while a subset is restricted to k raw indicators; this gap is the practical cost of requiring the final subset to be actual, interpretable indicators rather than statistical composites.

4.2.7 Headline Recommended Subset

The 21-indicator subset from flat allocation by sub-pillar above, with full indicator names:

Table 16: Headline subset: one reflective indicator per sub-pillar

Sub-Pillar	Indicator	r_m of Best Subset
Institutional environment	Operational stability for businesses*	0.938
Regulatory environment	Rule of law*	0.975
Business environment	Entrepreneurship policies and culture†	0.834
Education	School life expectancy	0.626
Tertiary education	Tertiary enrolment	0.615
Research and development (R&D)	Gross expenditure on R&D	0.793
Information and communication technologies (ICTs)	ICT access*	0.877
General infrastructure	Logistics performance*	0.647
Ecological sustainability	ISO 14001 environmental certificates	0.588
Credit	Finance for startups and scale-ups†	0.650
Investment	Joint venture deals	0.750
Trade, diversification and market scale	Domestic industry diversification	0.639
Knowledge workers	Knowledge-intensive employment	0.802

Sub-Pillar	Indicator	r_m of Best Subset
Innovation linkages	University–industry R&D collaboration†	0.727
Knowledge absorption	Research talent	0.508
Knowledge creation	PCT patents by inventor origin	0.638
Knowledge impact	High-tech manufacturing	0.592
Knowledge diffusion	Production and export complexity	0.649
Intangible assets	Industrial designs by origin	0.613
Creative goods and services	Entertainment and media market	0.550
Online creativity	GitHub commits	0.639

Finding: This subset captures r_m^2 68.2% of the full dataset’s PCA variance structure and recovers a Rand Index of 0.923 against the full-data country clustering, while cutting the indicator count from 78 to 21 (a 73% reduction).

5 Cross-Allocation Comparison

Table 17: Country perspective: all four allocation rules compared

Allocation Rule	Indicators	Rank Correlation
Flat allocation (pillar)	21	0.834
Flat allocation (sub-pillar)	21	0.844
Proportional allocation (pillar)	42	0.941
Proportional allocation (sub-pillar)	42	0.929

Table 18: Variable perspective: all four allocation rules compared

Allocation Rule	Indicators	r_m	r_m^2 (%)
Flat allocation (pillar)	21	0.854	72.9
Flat allocation (sub-pillar, headline)	21	0.826	68.2
Proportional allocation (pillar)	42	0.963	92.6
Proportional allocation (sub-pillar)	42	0.946	89.5

Finding: Across both perspectives and both metrics, proportional allocation at 42 indicators outperforms flat allocation at 21 indicators - but budget size, not the allocation rule itself,

explains most of that gap (see the random-baseline comparisons above, including the exact 42-indicator random baseline). Within a fixed budget, the two allocation rules perform comparably; the practical case for proportional allocation is representational (no pillar/sub-pillar is systematically over- or under-weighted relative to its actual size in the index), not a demonstrated statistical edge over flat allocation on these metrics alone.

6 Conclusions

1. **Meaningful reduction is possible, with a clear methodological trade-off.** The headline 21-indicator subset (one reflective indicator per sub-pillar) captures r_m^2 68.2% of the full PCA’s variance structure and a Rand Index of 0.923 against the full clustering, at a 73% reduction in indicator count.
2. **Exhaustive within-group search does not clearly outperform random selection at the 21-indicator budget**, for either metric: both the country-perspective and variable-perspective exhaustive strategies score close to what an unconstrained random subset of similar size averages. The gain from exhaustive search becomes more visible at the larger 42-indicator budget, though even there the margin over a random baseline of the same size is modest.
3. **Flat and proportional allocation perform comparably at a fixed budget**; proportional allocation’s main practical benefit is representational (no pillar/sub-pillar is systematically over- or under-weighted relative to its actual size in the index), not a demonstrated statistical edge over flat allocation at matched size.
4. **Rand Index is not monotonic with rank correlation or subset size** - the 42-indicator proportional subset scores a lower Rand Index than the 21-indicator flat subset despite a higher rank correlation, because SIMPROF’s significance-tested cluster boundaries can shift disproportionately with a different indicator mix.
5. **The r_m criterion treats “information” as PCA variance**, which may downweight indicators with low variance but high substantive/policy relevance; the headline subset should be reviewed against domain expertise, not adopted on statistical grounds alone.

7 Limitations

1. **Single-year analysis.** This analysis covers GII 2025 only; temporal stability of the recommended subset across years has not been assessed.
2. **PCA-defined “information.”** The r_m criterion equates information content with PCA variance structure; a substantively important but low-variance indicator could be excluded even if policy-relevant.

3. **Local search proxy.** Exhaustive search within each pillar/sub-pillar uses a local r_m computed on that group’s own sub-PCA (not the full 78-indicator PCA) as a tractable stand-in for an otherwise infeasible full search; the assembled subset is re-scored against the true full-dataset metric for reporting, but the *search* itself never directly optimizes that true metric.
4. **Linear relationships assumed.** Both the rank-correlation and r_m criteria are built on linear/monotonic structure (PCA, Euclidean distance); non-linear relationships between indicators are not captured.
5. **Domain-specific validity not tested.** Whether the recommended subset remains meaningful to GII’s intended policy audience has not been evaluated here and would require domain-expert review.

8 Data Availability

All data and code are available in this repository:

- Raw data: `gii-data/idata.xlsx`, `gii-data/imeta.xlsx`
- Processed data: `data/gii_data_standardized.csv`, `data/gii_data_complete_cases.csv`
- Analysis scripts: `scripts/00_setup.R` through `scripts/04_variable_perspective.R`, shared functions in `scripts/R/`
- Results: `outputs/` (CSVs for tabular results, `.rds` for clustering objects)

9 References

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10 Appendix

11 Data Imputation Documentation

Missing data is an inherent challenge in composite index construction. This analysis used k-Nearest Neighbors (kNN) imputation to address missing values before conducting the main analyses.

11.1 Imputation Process

- **Method:** k-Nearest Neighbors (kNN) imputation
- **Parameters:** k = 5 neighbors (VIM: :kNN, other parameters left at package defaults)
- **Implementation:** R VIM package
- **Indicators Used:** Level 1 indicators from IMETA metadata

11.2 Missing Data Statistics

Table 19: Imputation Process Summary

Metric	Count
Total Missing Values	1211
Variables with Missing Data	58
Countries with Missing Data	135

Note: 1211 missing values across 58 of 78 indicators, affecting 135 of 139 countries, were imputed prior to standardization.

12 Descriptive Statistics Tables

12.1 Descriptive Statistics - Original Data

The table below presents descriptive statistics for all Level 1 indicators in the original dataset before imputation:

Table 20: Descriptive Statistics - Original Data (n=78 indicators)

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Government effectiveness*	0.15	0.03	-1.75	2.32	-0.71	0.29
Operational stability for businesses*	2.23	2.15	0.70	4.45	0.03	0.58
Regulatory quality*	0.16	0.11	-2.03	2.31	-0.72	0.21
Rule of law*	0.07	-0.12	-2.15	1.97	-0.77	0.29
Entrepreneurship policies and culture†	4.72	4.72	2.92	7.25	-0.21	0.45
Policy stability for doing business†	4.03	3.98	1.83	6.45	-0.69	0.02
Expenditure on education	4.25	4.16	0.35	9.39	0.80	0.29
Government funding/pupil, secondary	19.60	19.41	6.00	51.06	1.57	0.74
PISA scales in reading, maths and science	439.44	439.29	337.45	579.03	-0.84	0.07
Pupil-teacher ratio, secondary	15.89	13.53	6.14	68.13	9.84	2.46
School life expectancy	14.17	14.36	6.34	20.85	-0.23	-0.28
Graduates in science and engineering	23.29	23.14	9.56	41.10	-0.33	0.17
Tertiary enrolment	51.51	54.40	2.71	166.67	-0.02	0.37
Tertiary inbound mobility	7.34	4.24	0.00	69.73	15.40	3.35
QS university ranking*	21.29	10.43	0.00	97.63	0.29	1.14
Global corporate R&D investors	926.35	0.00	0.00	36142.79	60.93	7.11
Gross expenditure on R&D	1.01	0.61	0.01	6.35	4.50	1.92
Researchers	2269.70	922.47	4.50	10413.00	0.53	1.21
Government's online service*	0.68	0.72	0.17	1.00	-0.95	-0.40
ICT access*	83.02	88.25	24.33	100.00	0.54	-1.12
ICT use*	79.65	84.47	29.84	99.36	1.78	-1.49

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Electricity output	4936.33	3183.45	29.76	53263.16	24.28	4.06
Gross capital formation	24.49	23.70	11.60	44.56	0.63	0.74
Logistics performance*	3.12	3.10	2.10	4.30	-1.10	0.19
GDP/unit of energy use	12.23	11.06	2.16	41.21	4.01	1.59
ISO 14001 environmental certificates	2.05	0.99	0.04	11.87	2.95	1.82
Low-carbon energy use	22.00	18.38	0.01	82.64	0.82	1.07
Domestic credit to private sector	60.33	48.53	8.14	248.76	2.51	1.51
Finance for startups and scale-ups†	4.49	4.49	1.99	6.90	-0.48	0.09
Loans from microfinance institutions	2.09	1.05	0.00	17.34	12.83	3.33
Joint venture deals	0.21	0.06	0.00	4.55	56.21	6.66
Late VC deals intensity	0.32	0.03	0.00	11.43	55.80	7.13
Market capitalization	80.85	37.45	0.20	1507.77	55.74	6.96
Venture capital investors	0.65	0.14	0.01	29.23	101.26	9.79
Venture capital recipients	0.24	0.08	0.01	5.50	67.12	7.42
Applied tariff rate, weighted avg.	4.16	2.34	0.00	47.54	40.98	5.09
Domestic market scale	1370.77	253.12	4.14	37072.09	47.47	6.55
Domestic industry diversification	0.18	0.14	0.07	0.55	3.07	1.61
Emerging knowledge workforce	39.00	35.41	17.63	66.87	-0.99	0.53
Females employed w/advanced degrees	13.49	12.78	0.14	31.04	-1.14	0.26
GERD financed by business	34.31	38.44	0.00	80.84	-1.07	-0.07
GERD performed by business	0.77	0.38	0.00	5.90	7.67	2.33
Knowledge-intensive employment	27.07	23.94	2.02	68.01	-0.95	0.31
Patent families	1.30	0.03	0.00	73.67	114.09	10.30
Public research–industry co-publications	2.01	1.34	0.22	9.07	2.97	1.78
State of cluster development†	4.00	3.90	2.59	5.40	-0.79	0.24
WU Rankings industry and global outlook	55.74	49.40	26.70	97.10	-0.99	0.58
University–industry R&D collaboration†	3.97	3.86	2.30	6.72	-0.12	0.38
FDI net inflows	3.93	2.50	- 161.34	224.25	66.45	3.27
High-tech imports	9.33	8.01	1.30	55.19	21.43	3.65
ICT services imports	1.75	1.44	0.01	16.01	31.07	4.44
Intellectual property payments	1.05	0.54	0.00	23.25	64.28	7.17
Research talent	33.25	32.27	0.30	81.91	-1.19	0.21

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Citable documents H-index	466.22	281.00	53.00	3213.00	7.98	2.39
Patents by origin	2.54	0.62	0.00	61.56	39.40	5.92
PCT patents by inventor origin	0.62	0.07	0.00	7.57	11.91	3.30
Scientific and technical articles	13.42	9.94	0.55	47.72	0.52	1.13
Utility models by origin	1.55	0.16	0.00	88.51	84.62	9.17
Software spending	0.24	0.21	0.00	1.26	3.08	1.42
Labor productivity growth	0.83	0.86	-4.57	6.26	0.84	-0.01
High-tech manufacturing	25.19	21.99	1.04	79.65	0.15	0.75
Unicorn valuation	1.09	0.00	0.00	21.85	27.52	4.75
Production and export complexity	0.09	-0.05	-2.18	2.29	-0.52	0.23
High-tech exports	4.57	1.51	0.02	48.39	12.41	3.08
ICT services exports	2.99	1.68	0.00	36.98	26.94	4.55
Intellectual property receipts	0.48	0.07	0.00	5.36	9.15	3.01
ISO 9001 quality certificates	5.46	3.23	0.04	29.87	3.42	1.80
Global brand value, top 5,000	3.59	0.78	0.00	21.12	1.78	1.64
Intangible asset intensity, top 15	41.66	56.72	-	89.79	28.96	-4.81
			371.30			
Industrial designs by origin	2.21	0.92	0.00	23.28	11.87	3.25
Trademarks by origin	36.72	28.95	0.00	202.33	7.36	2.24
Creative goods exports	1.27	0.38	0.00	11.50	7.00	2.60
Cultural and creative services exports	0.96	0.46	0.00	18.31	53.87	6.48
Entertainment and media market	0.93	0.47	0.02	3.59	-0.06	0.92
National feature films	5.01	2.67	0.00	46.93	18.95	3.70
Mobile app creation	1379627.73	181351.44	0.04	98744342.48	123.62	10.98
GitHub commits	1326116.27	410927.87	2571.55	14402260.97	14.47	2.99
Top-level domains (TLDs)	117.16	22.31	0.12	2538.57	46.17	5.93

12.2 Descriptive Statistics - Complete Data

The table below presents descriptive statistics for all Level 1 indicators in the complete dataset after kNN imputation and case-wise cleaning:

Table 21: Descriptive Statistics - Complete Data (n=78 indicators)

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Operational stability for businesses*	2.23	2.15	0.70	4.45	0.03	0.58
Government effectiveness*	0.15	0.03	-1.75	2.32	-0.71	0.29
Regulatory quality*	0.16	0.11	-2.03	2.31	-0.72	0.21
Rule of law*	0.07	-0.12	-2.15	1.97	-0.77	0.29
Policy stability for doing business†	4.03	3.98	1.83	6.45	-0.65	0.02
Entrepreneurship policies and culture†	4.63	4.54	2.92	7.25	0.28	0.64
Expenditure on education	4.24	4.14	0.35	9.39	0.83	0.31
Government funding/pupil, secondary	18.48	17.96	6.00	51.06	1.96	0.85
School life expectancy	13.92	13.94	6.34	20.85	-0.42	-0.13
PISA scales in reading, maths and science	419.11	403.60	337.45	579.03	-0.43	0.70
Pupil-teacher ratio, secondary	15.80	13.28	6.14	68.13	10.01	2.48
Tertiary enrolment	50.58	53.42	2.71	166.67	0.00	0.41
Graduates in science and engineering	22.88	22.88	9.56	41.10	-0.31	0.25
Tertiary inbound mobility	6.61	3.14	0.00	69.73	17.76	3.58
Researchers	1922.18	669.77	4.50	10413.00	1.08	1.44
Gross expenditure on R&D	0.88	0.42	0.01	6.35	5.71	2.16
Global corporate R&D investors	933.55	0.00	0.00	36142.79	61.25	7.12
QS university ranking*	21.29	10.43	0.00	97.63	0.29	1.14
ICT access*	82.88	88.19	24.33	100.00	0.47	-1.09
ICT use*	78.81	83.81	29.84	99.36	1.28	-1.36
Government's online service*	0.68	0.73	0.17	1.00	-0.94	-0.41
Electricity output	4663.61	2879.19	29.76	53263.16	25.59	4.17
Logistics performance*	3.04	2.90	2.10	4.30	-0.96	0.45
Gross capital formation	24.48	23.70	11.60	44.56	0.67	0.75
GDP/unit of energy use	12.00	10.93	2.16	41.21	4.46	1.69
Low-carbon energy use	22.00	18.38	0.01	82.64	0.82	1.07
ISO 14001 environmental certificates	2.05	0.99	0.04	11.87	2.95	1.82
Finance for startups and scale-ups†	4.25	4.11	1.99	6.90	-0.45	0.48
Domestic credit to private sector	60.08	48.30	8.14	248.76	2.47	1.51

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Loans from microfinance institutions	1.43	0.63	0.00	17.34	26.25	4.59
Market capitalization	58.57	25.34	0.20	1507.77	88.80	8.73
Venture capital investors	0.56	0.11	0.01	29.23	120.91	10.68
Venture capital recipients	0.22	0.07	0.01	5.50	73.92	7.78
Late VC deals intensity	0.25	0.01	0.00	11.43	71.99	8.09
Joint venture deals	0.18	0.04	0.00	4.55	66.13	7.21
Applied tariff rate, weighted avg.	4.16	2.34	0.00	47.54	40.98	5.09
Domestic industry diversification	0.19	0.17	0.07	0.55	2.05	1.19
Domestic market scale	1370.77	253.12	4.14	37072.09	47.47	6.55
Emerging knowledge workforce	39.00	35.41	17.63	66.87	-0.99	0.53
Knowledge-intensive employment	26.67	23.31	2.02	68.01	-0.89	0.38
Females employed w/advanced degrees	13.06	11.20	0.14	31.04	-1.04	0.35
GERD performed by business	0.50	0.06	0.00	5.90	11.94	2.97
GERD financed by business	28.45	28.73	0.00	80.84	-1.02	0.34
Public research–industry co-publications	2.01	1.34	0.22	9.07	2.97	1.78
University–industry R&D collaboration†	3.95	3.83	2.30	6.72	-0.05	0.42
WU Rankings industry and global outlook	51.65	42.20	26.70	97.10	-0.25	1.02
State of cluster development†	3.99	3.84	2.59	5.40	-0.75	0.28
Patent families	1.29	0.03	0.00	73.67	114.91	10.33
Intellectual property payments	1.05	0.54	0.00	23.25	64.28	7.17
High-tech imports	9.32	8.03	1.30	55.19	21.61	3.67
ICT services imports	1.75	1.44	0.01	16.01	31.07	4.44
FDI net inflows	3.91	2.49	-	224.25	66.93	3.29
			161.34			
Research talent	23.49	11.34	0.30	81.91	-0.61	0.86
Patents by origin	2.51	0.61	0.00	61.56	39.97	5.96
PCT patents by inventor origin	0.58	0.06	0.00	7.57	13.22	3.45
Utility models by origin	1.05	0.16	0.00	88.51	136.59	11.64
Scientific and technical articles	13.42	9.94	0.55	47.72	0.52	1.13
Citable documents H-index	466.22	281.00	53.00	3213.00	7.98	2.39
Labor productivity growth	0.83	0.86	-4.57	6.26	0.87	0.00
Unicorn valuation	1.09	0.00	0.00	21.85	27.52	4.75
Software spending	0.24	0.20	0.00	1.26	3.09	1.42
High-tech manufacturing	22.20	18.35	1.04	79.65	0.37	0.94
Intellectual property receipts	0.48	0.07	0.00	5.36	9.15	3.01

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Production and export complexity	0.08	-0.06	-2.18	2.29	-0.43	0.26
High-tech exports	4.53	1.51	0.02	48.39	12.49	3.09
ICT services exports	2.99	1.68	0.00	36.98	26.94	4.55
ISO 9001 quality certificates	5.46	3.23	0.04	29.87	3.42	1.80
Intangible asset intensity, top 15	32.37	39.35	- 371.30	89.79	34.44	-4.68
Trademarks by origin	36.74	28.99	0.00	202.33	7.57	2.26
Global brand value, top 5,000	3.20	0.47	0.00	21.12	2.47	1.81
Industrial designs by origin	2.12	0.84	0.00	23.28	12.62	3.34
Cultural and creative services exports	0.86	0.42	0.00	18.31	60.72	6.87
National feature films	4.47	2.69	0.00	46.93	26.15	4.26
Entertainment and media market	0.54	0.17	0.02	3.59	2.74	1.84
Creative goods exports	1.27	0.38	0.00	11.50	7.07	2.61
Top-level domains (TLDs)	117.16	22.31	0.12	2538.57	46.17	5.93
GitHub commits	1357449.63	411718.75	2571.55	14402260.97	13.92	2.91
Mobile app creation	1305696.24	155386.88	0.04	98744342.48	131.12	11.31